OOP LAB 6

SOLVED

Walkthrough Task:

When we use more than one constructor in class it is called constructor overloading.

W T.cpp

```cpp
#include<iostream>
using namespace std;
class Rectangle{
	private:
		double length;
		double width;
	public:
		Rectangle() {
		length = 0;
		width = 0;
		}
		Rectangle(double l, double w){
			length = l;
			width = w;
		}
		Rectangle(const Rectangle &r){
			length = r.length;
			width = r.width;
		}
		double area(){
			return length * width;
		}
		~Rectangle(){
			cout<<"Destructor Called For Object"<<endl;
		}
};
int main()
{
	Rectangle r1;
	cout<<"Area = "<<r1.area()<<endl;
	Rectangle r2(14.9,23.67);
	cout<<"Area = "<<r2.area()<<endl;
	Rectangle r3 = r2;
	cout<<"Area = "<<r3.area()<<endl;

}
```

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```
Area = 0
Area = 352.683
Area = 352.683
Destructor Called For Object
Destructor Called For Object
Destructor Called For Object

--------------------------------
Process exited after 0.404 seconds with return value 0
Press any key to continue . . .
```

Practice Task 1:

W T.cpp P T 1.cpp

```cpp
#include<iostream>
using namespace std;
class Student{
    private:
        string name;
        int rollnumber;
    public:
        Student(){
            name = "Unknown";
            rollnumber = 0;
            cout<<"Default Constructor Called"<<endl;
        }
        Student(string n, int r){
            name = n;
            rollnumber = r;
            cout<<"Parameterized Constructor Called"<<endl;
        }
        Student(const Student &s){
            name = s.name;
            rollnumber = s.rollnumber;
            cout<<"Copy Constructor Called"<<endl;
        }
        ~Student(){
            cout<<"Destructor Called For Object: "<<name<<endl;
        }
        void display(){
            cout<<endl;
            cout<<"Name: "<<name<<endl;
            cout<<"Roll Number: "<<rollnumber<<endl;
            cout<<"\n=========================\n";
        }
};
int main()
{
    Student s1;
    Student s2("Ali",45);
    Student s3("Hamza",12);
    Student s4("Usman",56);
    Student s5("Umar",22);
    Student s6 = s5;
    s1.display();
    s2.display();
    s3.display();
    s4.display();
    s5.display();
    s6.display();
    return 0;
}
```

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```
Default Constructor Called
Parameterized Constructor Called
Parameterized Constructor Called
Parameterized Constructor Called
Parameterized Constructor Called
Copy Constructor Called

Name: Unknown
Roll Number: 0

=========================

Name: Ali
Roll Number: 45

=========================

Name: Hamza
Roll Number: 12

=========================

Name: Usman
Roll Number: 56

=========================

Name: Umar
Roll Number: 22

=========================

Name: Umar
Roll Number: 22

=========================
Destructor Called For Object: Umar
Destructor Called For Object: Umar
Destructor Called For Object: Usman
Destructor Called For Object: Hamza
Destructor Called For Object: Ali
Destructor Called For Object: Unknown

--------------------------------
Process exited after 0.336 seconds with return value 0
Press any key to continue . . .
```

Practice Task 2:

W T.cpp P T 1.cpp P T 2.cpp

```cpp
#include<iostream>
using namespace std;
class BankAccount{
    private:
        int accountNumber;
        double balance;
    public:
        BankAccount(){
            accountNumber = 0;
            balance = 0;
        }
        BankAccount(int a, double b){
            accountNumber = a;
            balance = b;
        }
        void deposit(double amount){
            balance = balance + amount;
            cout<<"Balance After Deposit: "<<balance<<endl;
        }
        void withdraw(double amount){
            if(amount <= balance){
            balance = balance - amount;
            cout<<"Balance After Withdraw: "<<balance<<endl;
        }
        else{
            cout<<"Insufficient Balance"<<endl;
        }
        }
        BankAccount(const BankAccount &b){
            accountNumber = b.accountNumber;
            balance = b.balance;
        }
        ~BankAccount(){
            cout<<"Accont Closed: "<<accountNumber<<endl;
        }
};
int main()
{
    BankAccount b1;
    b1.deposit(100);
    b1.withdraw(50);
    BankAccount b2(123,5000);
    b2.deposit(2000);
    b2.withdraw(1000);
    BankAccount b3(999,1000);
    b3.deposit(3000);
    b3.withdraw(1500);
```

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```
Balance After Deposit: 100
Balance After Withdraw: 50
Balance After Deposit: 7000
Balance After Withdraw: 6000
Balance After Deposit: 4000
Balance After Withdraw: 2500
Balance After Deposit: 6500
Balance After Withdraw: 4500
Balance After Deposit: 2000
Balance After Withdraw: 0
Balance After Deposit: 2500
Insufficient Balance
Balance After Deposit: 4500
Balance After Withdraw: 3000
Accont Closed: 523
Accont Closed: 523
Accont Closed: 253
Accont Closed: 452
Accont Closed: 999
Accont Closed: 123
Accont Closed: 0

--------------------------------
Process exited after 0.4419 seconds with return value 0
Press any key to continue . . .
```

W T.cpp | P T 1.cpp | P T 2.cpp

```cpp
                cout<<"Balance After Withdraw: "<<balance<<endl;
            }
            else{
                cout<<"Insufficient Balance"<<endl;
            }
            }
            BankAccount(const BankAccount &b){
                accountNumber = b.accountNumber;
                balance = b.balance;
            }
            ~BankAccount(){
                cout<<"Accont Closed: "<<accountNumber<<endl;
            }
};
int main()
{
    BankAccount b1;
    b1.deposit(100);
    b1.withdraw(50);
    BankAccount b2(123,5000);
    b2.deposit(2000);
    b2.withdraw(1000);
    BankAccount b3(999,1000);
    b3.deposit(3000);
    b3.withdraw(1500);
    BankAccount b4(452,2500);
    b4.deposit(4000);
    b4.withdraw(2000);
    BankAccount b5(253,1000);
    b5.deposit(1000);
    b5.withdraw(2000);
    BankAccount b6(523,1500);
    b6.deposit(1000);
    b6.withdraw(3000);
    BankAccount b7 = b6;            // Copy constructor copies only data (accountNumber, balance)
    b7.deposit(2000);               // It does NOT copy function calls like deposit() or withdraw()
    b7.withdraw(1500);              // After copying, b7 is a separate object – it does not follow b6 automatically
    return 0;                       // To change b7, I must call functions again and pass values manually
                                    // e.g., b7.deposit(500);  b7.withdraw(200);
}
```

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```
Balance After Deposit: 100
Balance After Withdraw: 50
Balance After Deposit: 7000
Balance After Withdraw: 6000
Balance After Deposit: 4000
Balance After Withdraw: 2500
Balance After Deposit: 6500
Balance After Withdraw: 4500
Balance After Deposit: 2000
Balance After Withdraw: 0
Balance After Deposit: 2500
Insufficient Balance
Balance After Deposit: 4500
Balance After Withdraw: 3000
Accont Closed: 523
Accont Closed: 523
Accont Closed: 253
Accont Closed: 452
Accont Closed: 999
Accont Closed: 123
Accont Closed: 0

--------------------------------
```

Practice Task 3:

W T.cpp | P T 1.cpp | P T 2.cpp | P T 3.cpp

```cpp
#include<iostream>
using namespace std;
class Product{
    private:
        string productName;
        double price;
        int quantity;
    public:
        Product(){
            productName = "Unknown";
            price = 0.0;
            quantity = 0;
        }
        Product(string pn, double p, int q){
            productName = pn;
            price = p;
            quantity = q;
        }
        Product(const Product &p){
            productName = p.productName;
            price = p.price;
            quantity = p.quantity;
        }
        void display(){
            cout<<"Product Name: "<<productName<<endl;
            cout<<"Product Price: "<<price<<endl;
            cout<<"Product Quantity: "<<quantity<<endl;
            cout<<"========================"<<endl;
        }
        ~Product(){
            cout<<"Product "<<productName<<" removed from the inventory"<<endl;
        }
};
int main()
{
    Product p1;
    p1.display();
    Product p2("Laptop",500000,5);
    p2.display();
    Product p3("Phone",80000,10);
    p3.display();
    Product p4("Watch",5000,25);
    p4.display();
    Product p5 = p4;
    p5.display();
    return 0;
}
```

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```
Product Name: Unknown
Product Price: 0
Product Quantity: 0
========================
Product Name: Laptop
Product Price: 500000
Product Quantity: 5
========================
Product Name: Phone
Product Price: 80000
Product Quantity: 10
========================
Product Name: Watch
Product Price: 5000
Product Quantity: 25
========================
Product Name: Watch
Product Price: 5000
Product Quantity: 25
========================
Product Watch removed from the inventory
Product Watch removed from the inventory
Product Phone removed from the inventory
Product Laptop removed from the inventory
Product Unknown removed from the inventory

--------------------------------
Process exited after 0.4177 seconds with return value 0
Press any key to continue . . .
```